

Box 27-1 Characteristics of Stress-Strain Behavior of Collagen

Stress and strain are linear up to the yield point. Elongation along this portion of the curve will not be permanent.

Elastic limit is the point to which maximum stress may be applied without permanent deformation.

Increased loading requires greater force to increase elongation of collagen because the tissue becomes stiffer.

Microfailure begins to occur in some of the fiber bundles in the linear region. However, the tissue retains its external appearance of continuity.

The end of the linear region is known as the *yield point of the tissue*. At this point, elongation of collagen occurs with a minimal increase in load. Beyond this point, major fiber bundle failure occurs, the external appearance changes, and the smooth outline is lost.

When maximum load point is reached ($P_{\max}$), complete rupture occurs.

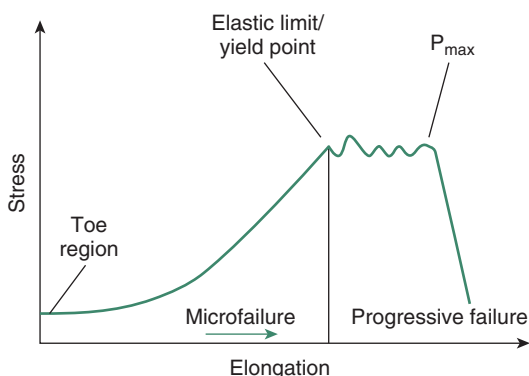


Figure 27-1 Typical stress-strain curve for connective tissue. Stress, Force applied per unit area; strain, proportional elongation; $P_{\max}$, maximum load point.

massage is undesirable because of increased tissue stiffness as a result of their viscoelastic properties. Also, dry tissue results in less compliance and elongates less readily. These are important facts for deciding the rate of massage application.

Lymphatics

In humans, one tenth of the tissue fluid is removed by the lymphatics. Understanding lymphatic anatomy, physiology, and function is necessary for successful massage. The rate of flow is determined by the interstitial fluid pressure and lymphatic pump activity.

Increased lymphatic flow is caused by the following:

- Increased capillary pressure
- Increased plasma osmotic pressure
- Increased interstitial fluid pressure
- Increased capillary permeability

Massage has been shown to increase lymph flow rate by 22-fold, lymph colloid concentration by 47 times, and leukocyte counts by 8.5 times.¹ It is believed that when tissues and fluid are compressed against the outer surfaces of the larger lymph vessels, lymph flow is impeded. This results in an increase in interstitial fluid pressure slightly above atmospheric pressure. Massaging may increase interstitial fluid pressure and help to move fluid into the lymphatic system. By massaging in a distal to proximal direction, it is proposed that fluid is moved from the extremity toward the central body core. This supports the practice of applying massage in a distal to proximal direction. Other controlled studies show that massage does not improve lymph drainage in certain situations, such as cancer. A study of women with breast cancer-related lymphedema evaluated the effect of compression garments with or without manual lymph drainage (massage). It was concluded that compression garments improved lymphedema, but there was no additional benefit of massage in relieving limb edema in women with breast cancer.²

Circulatory Effects

The pressure of the massage itself increases pressure within the tissues. Pressure gradients are created between the tissue spaces and vessels. As the hands are moved, changes in tissue pressure occur, creating fluctuating pressure differences between one area of tissue and another. It is believed that fluid moves constantly from tissues to vessels and back again, as it flows from areas of high pressure to areas of low pressure.

During an actual massage session, various massage manipulations are employed that are complex and involve a combination of squeezing, stretching, pulling, and traction forces, with movement occurring in different directions and in different tissue planes. Therefore there will be complex repetitive pressure changes occurring in varying directions and at different tissue depths. This is likely to have an effect on fluid interchange, whereby fluid is pushed from the tissue spaces into the vessels, toward the lymph nodes and heart, and new fluid is pushed or drawn into the spaces. Massage is believed to replenish the fluid in the spaces, producing a flushing effect, which brings in additional nutrients. There is evidence in humans that chemical irritants in the tissues (e.g., substance P, prostaglandins) and waste products of metabolism may decrease pain threshold by sensitizing free nerve endings. Replenishing tissue fluid and removing inflammatory products may reduce this effect, preventing or reducing some types of chronic pain. By removing metabolites and chemicals from muscles and “flushing” the muscle tissue with new circulation, muscle soreness following exercise may be reduced. However, controlled studies have not been able to demonstrate that massage reduces metabolites following exercise.^{3,4} Studies involving Doppler ultrasound have failed to demonstrate an effect of massage on muscle blood flow.^{5,6}

Tissue Movement

As the hands move along the superficial tissues, pressure to the tissues increases, and layers of tissue are moved. A light glide causes movement of the epidermis. If friction is maintained between the therapist's hand and the patient's skin, the epidermis moves with the hand and is gently stretched. There will also be some movement of the dermis because of the traction between the two layers. More pressure with friction, although still very light, results in traction between the dermis and the subcutaneous tissues. The end-feel of the stroke is when this traction reaches its limit and all layers are stretched.

Massage involves the interaction of energy between the person applying the massage and the dog, and the effects of touch to induce relaxation, communication, and a sense of general well-being. It may also produce movement of the tissues in subsequent layers as a result of traction of the tissue interfaces. Sensory and autonomic nerves may be stimulated, inducing changes in the nervous and circulatory systems, and movement may be affected in abnormal tissue, such as scar tissue or layers of adherent tissues. Reed and Held⁷ found no consistent systemic effect on autonomic nervous system parameters either acutely or cumulatively when comparing connective tissue massage and a placebo treatment (sham ultrasound) in healthy middle-aged and geriatric humans.

Therapeutic Effects

Massage is generally enjoyed by small animal patients and helps to relieve distress, anxiety, and discomfort. There are many postulated effects of massage reported in the literature. Unfortunately, many of these have not been substantiated by systematic scientific investigations.

Stroking young animals results in a reduction in the animal's physiologic response to stress, demonstrated by a decreased output of adrenocorticotrophic hormone.⁸ Young animals that are handled also show greater development of the cortex and the subcortex of the brain. They learn faster and have a more advanced stage of neural development than nonhandled animals.⁹ Resistance to infection later in life may also be beneficially influenced by cutaneous stimulation experienced by the infant animal.¹⁰

If a hand is placed over the surface of a dog's skin, heat is felt between the two surfaces. If the hand is placed on the skin and held still, this heat increases. Rubbing over the surface of the skin causes friction, and this results in an even greater increase in heat. Heat is a form of energy, and some schools of massage, particularly those grounded in Eastern practices, use the energy field that exists around the body. This is sometimes referred to as an *electromagnetic radiation* around the entire body. It is felt as heat and can be felt a small distance from the body. Some also believe that other forms of energy fields run through the body, along specific pathways known as

meridians. These are used in acupuncture and reflex therapy treatments.

Most of the research on the effects of massage has been done in humans. Most studies have evaluated healthy individuals, which may be misleading because physiologic compensatory mechanisms are extremely efficient in healthy tissues and any alterations in local blood flow may be compensated for by autoregulatory processes. Most of the research studies have also been performed on small groups and often without control groups. One study by Hansen and Kirstensen¹¹ gave little detail concerning the massage itself, such as the amount of pressure applied. Although the research supported the concept that massage increases blood flow in muscles, the lack of information regarding the type of massage and length of treatment make interpretation of the results difficult.

Changes in blood constituents following massage have provided information regarding the mechanism of action of massage.¹² Ernst¹³ found that a standard 20-minute massage treatment reduced the hematocrit and blood and plasma viscosity. It is believed that the fluid immediately surrounding poorly perfused vessels has low viscosity because of decreased cellularity and that the vasodilatation caused by massage aids in the recruitment of dormant vessels. It is desirable to increase local circulation to promote healing. The mechanical effect of the massage helps to remove low-viscosity tissue fluid into the circulation. The changes in blood detected within the vessels in this study support the belief that massage produces a flushing and mechanical effect on the circulation. In a more recent study, Hinds and colleagues⁶ showed no difference in femoral arterial blood flow, quadriceps muscle temperature, blood lactate levels, heart rate, or blood pressure when comparing post quadriceps exercise and massage to exercise and rest. They did show that skin blood flow and skin temperature were both significantly elevated in the massage group. They hypothesized that this potential diversion of blood flow away from the muscle could have a deleterious effect to recovering muscle tissue. They did not demonstrate a decrease in muscle blood flow, however, so this hypothesis is not supported by their work.

In a Cochrane database review, Furlan and colleagues¹⁴ critiqued 13 randomized trials and concluded that massage might be beneficial in the treatment of low back pain in acute and chronic cases, especially when combined with exercise and education.

Effects on Muscle

Massage may also affect skeletal muscles. Muscles have a natural resting tone that may be increased in postural muscles by external influences, such as cold or stress. This occurs because of the interaction of the muscle spindle and the central nervous system. Stretching a muscle may stimulate the muscle spindle and cause a reflex muscle contraction while reflex inhibition of the antagonist occurs.

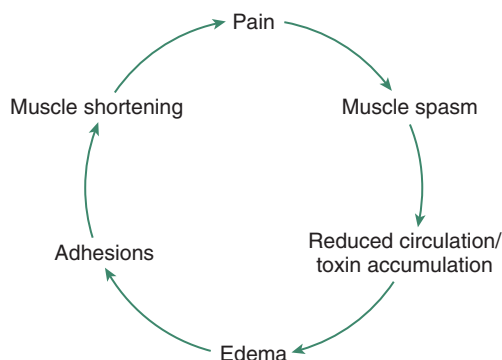


Figure 27-2 Self-perpetuating muscle spasm cycle.

Massage may add an external stimulus to sensory organs and either increases muscle tone by stimulation or reduces it, probably by facilitating an accommodation of the spindle, causing it to reset at a lower threshold of excitability.

It is believed that the circulatory effect of massage may reduce muscle soreness and therefore aid clinical signs associated with muscle injury or postexercise recovery. Danneskiold-Samsoe and colleagues¹⁵ studied the effects of massage for shoulder and back pain in humans. Blood samples were evaluated for myoglobin concentrations. There was a statistically significant effect, with peak myoglobin concentrations occurring within 3 hours of the massage. Myoglobin concentrations gradually decreased; after seven treatments, there was no difference. In a systematic review of the literature published in 1998, Ernst¹⁶ concluded that most studies evaluating the effect of massage on delayed-onset muscle soreness were methodologically flawed, findings were inconsistent between studies, and, although there appeared to be some potential for a positive effect, the evidence was not convincing and warranted further controlled investigation.

Jacobs¹⁷ advocated the use of massage for the relief of spasm of unknown cause but which resulted in a self-perpetuating muscle spasm cycle (Figure 27-2). Massage is also said to reduce tone in muscles that are tense or in spasm and to reduce the soreness and tenderness of exercised muscles. It is also used to prepare muscles for exercise. Hernandez-Reif and colleagues¹⁸ found that children with cerebral palsy who received 30 minutes of massage twice weekly for 12 weeks had decreased muscle spasticity of the arms, decreased arm muscle and overall tone, increased hip extension range of motion (ROM), and improved fine and gross motor function when compared with children with cerebral palsy who received 30 minutes of reading twice weekly.

Jones and Round¹⁹ discussed the process by which an analgesic substance, possibly potassium, is released when muscles maintain excessive tone or static contraction for prolonged periods. Eccentric exercise of muscles, which generates high forces in muscles, produces delayed-onset

soreness and a feeling of stiffness. Jones and Round¹⁹ explained this by identifying swelling in muscles and inflammation of the connective tissue. Massage may help reduce or prevent the initial tenderness. It is also believed that massage may help to prevent subsequent postexercise pain. Regular massages may make the muscles softer and more pliable, and the stretching manipulations may increase extensibility and strength in the connective tissues.¹⁵ Willems and colleagues²⁰ demonstrated decreased muscle soreness in portions of the quadriceps muscle (vastus lateralis and rectus femoris), but not in the vastus medialis after massage following eccentric exercise-induced muscle soreness. They also demonstrated improved vertical jumping performance in the massaged leg 48 hours after massage when compared with the nonmassaged leg. Mancinelli and colleagues²¹ showed significant effects of massage versus control in vertical jump displacement, perceived muscle soreness, and algometer muscle pain readings, but not timed shuttle run in women collegiate athletes.

Pain experienced after exercising with the muscles in a lengthened position may also be due to inflammation of the connective tissue rather than muscle damage. Massage may assist by increasing pliability in the connective tissue around the muscle fasciculi. Massage assists mobility between the interfaces within the muscle and also stretches any fibrous adhesions or scarring in the tissue.

The effects of massage on the Hoffman (H) reflex, which indicates the excitability in α -motoneurons or anterior horn cells, has also generated interest. Goldberg and colleagues²² studied the effects of massage on healthy people and patients with spinal cord injuries. They applied one-handed petrissage to deep and superficial tissues. There was a 49% reduction in H reflex activity with deep massage, and a 39% reduction with light massage. From this study, it was concluded that stimulating cutaneous mechanoreceptors and pressure receptors reduces H reflex activity. Sullivan and colleagues²³ also demonstrated a reduction in α -motoneuron excitability of massaged muscles.

Effects on Pain and Sensation

The effects of massage on connective tissues, which helps to reduce muscle spasm and soft tissue discomfort, have already been discussed. However, Ueda and colleagues²⁴ demonstrated that postsurgical massage may facilitate the regression of sensory analgesia following an epidural block.

Carreck²⁵ evaluated the effect of massage on levels of pain perception in humans. She found that pain threshold was raised to significantly higher levels in the massage group and this work suggests that massage may be used to help manage pain and reduce treatment soreness.

Massage may cause traction at tissue interfaces. Horizontal plexi lie at interfaces in the tissues and gentle pulling on these vessels may stimulate the accompanying

sympathetic nerves, which supply the mechanoreceptors. These receptors may be distorted by the massage, possibly lowering mechanoreceptor sensitivity and reducing pain and tenderness. If delayed-onset pain in the muscle is caused by connective tissue inflammation, the flushing effect in the surrounding fluids and removal of inflammatory mediators may increase the speed at which the inflammation resolves.

Effects of Massage on Connective Tissue

Following injury or disease, inflammation causes increased vascular permeability. The resulting edema contains fibrinogen, which may result in formation of fibrous tissue. Adhesions form within the tissues, binding tissue interfaces. These adhesions prevent normal sliding of tissues on each other and also reduce the ability of the fibers to spread apart. *Massage* is described as being able to stretch tissues that have become short, tight, or adhered. Petrissage massage stretches and pulls the tissues in various directions and mobilizes the adjacent connective tissue fibrils. The molecular cross-bridges between fibrils may be influenced and plastic changes may occur; their length may be maintained or increased because of elongation or creep. Mobility may also be increased at the fibrous cross-bridges between fibers and areas of adhesions by stretching connections between the fibers. This facilitates spreading and sliding at tissue interfaces, and longitudinal elongation helps to increase flexibility.

Following surgery, scars may form with adhesions of various tissue layers to the underlying tissues. Massage helps mobilize and soften adhesions by elongation, creep, and increased fluid exchange of tissues. Collagen fibers align themselves along lines of stress. Techniques that stretch the fibers and adhesions in different directions eventually restore their mobility, promoting remodeling along the lines of normal stresses.

Following injury, it is important to maintain local flexibility of the involved structures to assist overall suppleness and to reduce the possibility of reinjury resulting from decreased extensibility of normal and scar tissues. Massage should be used to promote the functional recovery of structures and prevent reinjury by maintaining mobility at the interfaces, both inside and outside the affected structures. Massage should be combined with controlled longitudinal stretching exercises.

One technique to mobilize scar tissue is deep-friction massage.²⁶ It is applied at 90 degrees to the fibers and delivers a stretch and separation between individual fibers. Deep friction is believed to create movement and induce reactive hyperemia,²⁷ stretch cross-bridges between fibers, and stretch adhesions within the structure. Fibrous scarring in tendons and their interfaces may become more pliable with massage. Although one researcher has suggested that deep friction application may result in some analgesia to the area,²⁸ there is little evidence to support this claim.

Although many theoretical effects of massage have been put forth, there is little high-quality research evidence to support these claims. A Cochrane review evaluating the effect of deep transverse friction massage for treating tendinitis found no evidence of benefit with its use and concluded that there was the need for a high-quality randomized study using validated outcome measures to further evaluate its usefulness.²⁹

How Massage Works

Massage may have immediate benefits to a patient, but it usually takes regular treatments to bring about significant improvements in a particular condition. Initially, massage may provide relief from pain, reduce tension, and help to sedate the nervous system. When pain has subsided, attempts to correct the underlying cause of the problem may begin. Healing is stimulated by both mechanical and reflex actions.

The mechanical effect of massage to the soft tissues may help to relieve tense muscles through relaxation, decreased pain, and increased mobility, and it may improve the circulation and lymphatic flow. The body, mind, and emotions all interact with the nervous system. It is possible that touching the skin sends messages, including reflex actions as the involuntary reaction of one part of the body to the stimulation of another part, to reduce the amount of stress hormones, which may lower blood pressure, slow breathing, improve digestion, and generate a sense of well-being. It is believed that massage also helps to release endorphins, the body's natural pain relievers, and to activate neuropeptides, the intercellular messages within the nervous system. These link the skin to the nervous system and the entire body, including the immune system. These actions as a result of massage are thought to have tremendous benefits.

Indications for Massage

Mechanical

If a dog has a chronic musculoskeletal problem, it may develop secondary problems, which lead to postural and gait adaptations. In time, these may also become primary problems in themselves as a result of changes in the soft tissues, which lead to unprotected joints, possible damage to the joints, and a painful response.

Surgical

If a dog has had surgery, massage may help to maintain mobility. If the dog must be confined for a period of rest or is on restricted exercise, it helps to maintain tone and muscle condition. Massage also helps psychologically by maintaining contact between the dog and the owner, and it may ease pain and discomfort. After resolution of the

surgical condition, massage is indicated to maintain flexibility of all joints and soft tissues, and to prevent further loss of function.

Disease

For dogs suffering a permanent low-grade condition, such as osteoarthritis, massage before exercise helps to encourage blood flow, delivery of nutrients, and maintenance of soft tissue extensibility. For those that suffer chronic imbalances, which have caused structural changes of the soft tissues, massage helps to relax and ease discomfort and may have a significant effect on tension and muscular effort.

Massage for the Competitive Dog

Proper massage applied to competitive dogs may help to establish or reestablish full tissue function. Massage may be employed during the training period for treatment and rehabilitation, and to enhance athletic performance. It may also help to prepare the dog for competition or exercise, and to reduce stress and aid relaxation. After hard exercise or competition, it aids recovery.

At an amateur level, training may sometimes be somewhat neglected, and massage can play a significant role in maintaining tissue pliability, in hopes of preventing injury. At the other end of the scale, top canine athletes are more likely to be overtrained, and massage can assist in recovery from fatigue. In this case, 2 hours should be allowed to elapse after competition before massage is started for maximum effects. Care must be exercised to not massage too deep and induce mechanical trauma, which may be detrimental to fatigued muscles.

Massage for Performance

To perform properly, the body must be able to adapt to the extra stresses placed on it. Requirements for excellence are power, agility, and coordination of the dog, and to attain this level of performance, the dog must cope with activity at the extremes of joint movement and muscle length. This places heavy demands on the strength and the endurance of the muscle, and the ability of the joints to move freely in all directions. The dog also needs freely mobile connective tissues. Inevitably, excess or repetitive stresses, or overwork, may increase the likelihood of other injuries, such as muscle strain. In all cases, this results in replacement of connective tissue with scar tissue, resulting in loss of power, flexibility, and movement in the tissues. Moreover, if the tissue is not rehabilitated before normal full activity is resumed, then it may never fully recover and return to normal limits. This in turn may overload the surrounding tissues, causing fatigue and hypertrophy of muscle groups.

Massage can be used to stretch and free this inelastic scar tissue and return tissue mobility to normal. It also helps to remodel and promote resorption of the scar and to reestablish full function of the tissues.

Fatigue

When a dog trains very hard, it may not fully recover between sessions, which may lead to general muscle soreness and increased likelihood of injury. Massage is used to decrease muscle tone and tension, thereby reducing muscular discomfort and promoting relaxation. There is also evidence to suggest that it helps recovery from fatigue, as compared with rest alone. It would therefore be very appropriate to use massage after competition, training, traveling, or any particularly stressful situation.

Muscle Soreness

Dogs may suffer soreness in their muscles as a result of extreme exertion and adverse environmental conditions. Endurance dogs galloping over uneven terrain in wet weather conditions or in extreme heat and humidity are examples of extreme conditions. This form of soreness normally begins after completion of the event, and sometimes not until the next day. Less fit animals may also suffer discomfort during such exercises, but as a result of metabolic insufficiency. Well-conditioned muscles generally do not suffer from this problem because they have a well-functioning capillary bed and adequate oxygenation of the blood supply. Obviously, a dog will be at a disadvantage if it has not had an opportunity to fully recover from the previous competitive efforts. An inadequate rest period may result in very severe discomfort caused by connective tissue inflammation, tenderness, and stiffness. Massage may delay the onset of soreness and reduce it.

Experience and some studies suggest that the ideal time to massage is 1 to 3 hours after exercise to enhance athletic performance and for the best therapeutic benefit. Also, for certain muscles, massage may enhance tissue lengthening more than stretching alone.

There are also contraindications to massage ([Box 27-2](#)).

Planning a Massage

When planning a massage for a competitive animal, the stage of training, the fitness of the animal, and the purpose of the massage must be considered. The various purposes of massage include relaxation, toning, precompetition warm up, intercompetition treatment, aiding muscle soreness 1 to 3 hours after competition, improving healing, relief of acute or chronic injury and fatigue, muscle soreness, and preventing adhesion formation. A thorough patient evaluation is necessary to assess areas that might benefit the most from proper massage techniques.

Patient Evaluation

During the initial observation period, care should be taken to ensure that the dog is standing on a flat, even surface in an area with adequate lighting.

Box 27-2 Contraindications to Massage

Do not massage if the dog is suffering from the following conditions:

- Shock (lowers blood pressure)
- Fever
- Acute inflammation (do not address area directly, but it would be acceptable to work elsewhere on the body)
- Skin problems (ringworm)
- Infectious diseases
- Acute stages of viral disease (herpes)



Figure 27-3 Caudal view of patient.

Caudal View

The patient is evaluated from a caudal view while standing. From this angle, the general posture is evaluated, looking for any skeletal abnormalities (e.g., scoliosis), areas of atrophy or hypertrophy, natural foot placement, weight-bearing tendencies of the limbs, head carriage, and the general appearance of the hindquarters and tail. The patient in [Figure 27-3](#) has arthritis. Note that the dog has a clamped tail and decreased weight bearing on the left hindlimb, with the limb held in slight flexion.

Cranial View

From the cranial view it is important to note the symmetry of the head, eyes, ears, and paws, head carriage, forelimb weight-bearing tendencies, and the natural position of the forelimbs (e.g., neutral, rotation, adduction, abduction). Note in [Figure 27-4](#) the deviation of the head to the left with slight rotation to the left and external rotation of the forelimbs. Also, the right paw is more splayed than the left.



Figure 27-4 Cranial view of patient.



Figure 27-5 Lateral view of patient.

Lateral View

It is important to observe both sides of the patient. Important points to note are the general posture, appearance of the muscles, head carriage, tail position, and limb joint angles. In the dog in [Figure 27-5](#), note the roached back and clamped tail. Also note that the limb angles cannot be truly assessed because the dog is not standing squarely and symmetrically.

Gait Assessment

Walk: Caudal View

At this stage it is important to assess any hindlimb lameness. The ability of the dog to walk in a straight line should be assessed, along with the fluency of hindlimb movement



Figure 27-6 Gait assessment at a walk—caudal view.



Figure 27-7 Gait assessment at a walk—cranial view.

(i.e., does the dog move forward in a straight line, or does it circumduct, abduct, adduct, and so forth). It is important that the dog handler moves in a straight line and is educated in handling of dogs for evaluation purposes. Observations should be made of the lumbosacral rhythm, or the rise and fall of the tuber coxae and tuber sacrale, and the presence of any protective muscle splinting that may be occurring. In the dog in [Figure 27-6](#), the tail is held low, the hindlimb toe-off phase initiates in an adducted position, the dog appears to move toward the left, and the dog appears to be dragging the left hind toe.

Walk: Cranial View

The cranial view should evaluate forelimb lameness and fluency of movement along with natural head carriage and foot placement. The nose, head, and neck deviate to the left, and foot placement appears close in the forelimbs in the dog depicted in [Figure 27-7](#).

Trot: Caudal View

Similar to assessing walking from a caudal position, it is also important to note any hindlimb lameness, symmetry, fluency of movement, protective splinting of muscle groups (e.g., hamstrings or gluteals), tail carriage, and limb carriage and placement at a trot ([Figure 27-8](#)).

Trot: Cranial View

As with observation at a walk, observe the dog for any lameness, head and neck carriage, forelimb carriage and foot placement, and symmetry and fluency of movement. In [Figure 27-9](#), note the low head carriage and the adducted position of the right forelimb.

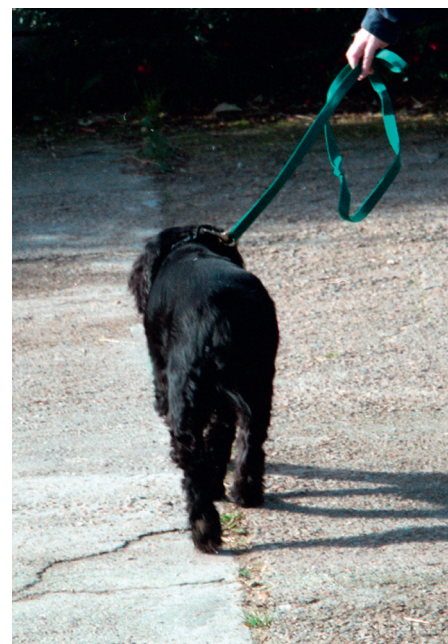


Figure 27-8 Gait assessment at a trot—caudal view.

Side View

Both sides of the animal should be assessed. From this angle it is important to note the equality of the hindlimb and forelimb stride lengths, the ability of all limbs to retract and protract, head and tail carriage, protective muscle splinting, and general muscle interaction. In [Figure 27-10](#), note the clamped appearance of the tail and possible guarding of the epaxial muscles.



Figure 27-9 Gait assessment at a trot—cranial view.



Figure 27-10 Gait assessment—side view.

Turning

It is important to evaluate the dog while turning in small circles in both directions, noting whether the bend follows through from nose to tail, whether the limbs cross over, and whether any areas look restricted. It is also important to recognize whether the dog actually understands what is being asked of it. In [Figure 27-11](#) note the reluctance of the dog to perform the task: the dog appears to be braced from neck to tail, and the dog is also not crossing his legs.

Palpation

The dog's skeletal structure and muscle tone should be assessed after evaluation of the dog's gait. This is performed by palpating along the dog, from the wings of the atlas, along the spine and limbs, to the tail ([Figure 27-12](#)). Areas of irritability, spasm, atrophy, hypertrophy, and asymmetry are noted, along with any skeletal abnormalities.



Figure 27-11 Turning short in small circles.



Figure 27-12 Palpation of the dog.



Figure 27-13 Palpation of dorsal spinous processes.

By palpating on either side of the dorsal spinous processes with the thumbs, sensitive areas may be assessed ([Figure 27-13](#)). The surrounding areas are also palpated for any evidence of any other sensitive areas in the region, such as the withers and triceps region ([Figure 27-14](#)). Limb musculature is also assessed independently (e.g., shoulder



Figure 27-14 Palpation of the scapular and triceps region.



Figure 27-16 Flexion of the cervical spine.



Figure 27-15 Palpation of the tuber sacrale and tuber coxae.

girdle and pelvic girdle musculature), noting whether any tone or spasm are present. Bony landmarks are palpated bilaterally, noting the symmetry of position and surrounding musculature, for example, the tuber sacrale and tuber coxae (Figure 27-15).

Passive Movements of the Cervical Spine

Flexion

While standing above the dog, place one hand over the cervical spine to enable assessment of the muscles and joints as they move through a ROM, then place the other hand over the dog's nose and guide the head down toward the forelimbs. Note the quality and range of movement and end feel, as well as the feel of the muscles and joints under the palpating hand. Although it is difficult to appreciate in the photograph, the dog in Figure 27-16 has restriction in flexion.

Extension

Support the dog's head under its chin with one hand and palpate the cervical spine with the other. Using the



Figure 27-17 Extension of the cervical spine.

supporting hand, guide the head dorsally into extension. Note the quality and range of movement, the end feel, and the characteristics of the muscles and joints under the palpating hand. The dog in Figure 27-17 appears to resist this maneuver.

Side Flexion

While placing one hand over one side of the dog's nose, use the other hand to stabilize the dog while palpating the cervical spine. First guide the dog's head to one direction, and then after changing positions of the hands, guide the head in the other direction. Note the range and quality of movement, end feel, and so forth for both sides. In Figure 27-18, left-side flexion appears restricted, and increased tone is palpated by the evaluator in the cervical musculature.



Figure 27-18 Side flexion of the cervical spine.



Figure 27-19 Flexion of a forelimb.

Forelimb

Flexion

While ensuring that the dog is standing supported with the head pointing forward, flex the carpus and elbow. Note the range of movement, quality of movement, end feel, and scapular elevation. The dog in Figure 27-19 has good flexion in these joints.

Protraction

After evaluating flexion, continue to guide the limb forward into full protraction. Again, note the range and quality of movement and the end feel. Any areas of increased muscle



Figure 27-20 Protraction of a forelimb.



Figure 27-21 Retraction of a forelimb.

tone should be identified. Moving the limb across the midline into adduction tests the latissimus dorsi muscle. In this dog, protraction of the limb appears to be adequate (Figure 27-20).

Retraction

From the protracted position, guide the limb back into retraction, ensuring that the dog's neck is not tilted from midline. The range, quality, and end feel of movement are noted. The dog in Figure 27-21 appears to have restricted movement, possibly because of increased tone in the brachiocephalicus muscle, and the dog does not appear to be comfortable.

Cervical Spine Side Flexion with Forelimb Retraction

Support the dog between your legs and with one hand, flex the neck to the side. Hold this position, and use the other hand to gently retract the opposite forelimb to assess the range of movement (Figure 27-22).



Figure 27-22 Side flexion of the cervical spine with retraction of the opposite forelimb.



Figure 27-23 Adduction of a forelimb.

Adduction

From the neutral position, guide the limb across the chest toward the opposite leg. Note the range, quality, and end feel of the movement (Figure 27-23).

Hindlimb

Flexion

With the dog supported, flex the hindlimb, obtaining full flexion of the hock and stifle. Assess the range and quality of movement, end feel, and the muscles. The dog in Figure 27-24 appears to have decreased limb flexion. Also note the dog's stance in the other limbs and overall level of comfort.

Protraction

With one hand palpating the hindquarters and caudal thigh muscles, allow the other hand to guide the limb forward into protraction. Assess the range and quality of movement, end feel, and muscles. The dog in Figure 27-25 appears to have full protraction, but may also have compensatory flexion of the lumbosacral area.



Figure 27-24 Flexion of a hindlimb.



Figure 27-25 Protraction of a hindlimb.

Retraction

With the dog in a standing position, use one hand to support the limb and guide it into retraction. Use the other hand to palpate the quadriceps and caudal thigh muscles. Again, note the range and quality of movement, end feel, and any fasciculation of the muscle. The dog in Figure 27-26 has very restricted retraction.

Note: All of the passive limb movements described previously can also be performed with the dog in lateral recumbency.

Tail

Tail movements may also be assessed, noting range of movement, quality of movement, and end feel. Elevation, depression, side flexion, and rotation movement of the tail can all be assessed (Figure 27-27).

Practical Application of Massage

Body Position and Comfort of the Clinician

Performing a massage can be quite tiring initially because you are concentrating on a new task and your body is



Figure 27-26 Retraction of a hindlimb.



Figure 27-27 Assessment of tail motion.

learning new movements. If the clinician is not careful, this may lead to tension and fatigue. To prevent this from occurring, try to work in a systematic way. Also, be careful to maintain proper body posture and relax while learning the moves to help minimize these risks. Using your larger muscles such as the shoulder muscles instead of your hands helps to minimize fatigue. It is important that the dog feel confident in you, too. Dogs appear to be very good at picking up tension and any lack of confidence.

Maintaining good posture means greater comfort, and the clinician will not fatigue as quickly. As a result, the massage will be more enjoyable for the clinician, resulting in greater relaxation for both parties. Altogether, the massage will be more effective, and a better result will be achieved.

Identification of Massage Areas and Application of Massage

When massaging, you may work on the entire dog, or selected areas. If the dog has a diagnosed condition, local

massage may be appropriate on a regular basis. General massage helps with treatment of compensatory conditions and biomechanical changes.

Massage may be performed to isolated areas or to the entire limb and body. It may be used in conjunction with other therapies and veterinary treatments. It is also useful to be applied by the owner and may be successfully taught to the owner by the therapist.

Massage Techniques

Massage has many benefits for dogs, and there are many techniques that may be employed. Massage involves the use of a series of movements using the hands, which have an effect on the tissues. It is important to comfortably position the dog in lateral recumbency. It is also important to position yourself comfortably. The person performing the massage should be beside the dog on the floor, or place the dog on a raised surface, which may prevent you from stooping over or compromising your posture. It is best to start at one end, such as the head, and work along the dog in a sequenced manner (e.g., head, neck, shoulders, forelimbs, back, hindquarters, and tail). The following are techniques that may be used at any stage and over any muscle group.

Stroking

Stroking is a good technique to start the treatment ([Box 27-3](#)). When the dog is positioned, run your hand over the dog from neck to tail and down the limbs with medium pressure. This technique aids relaxation of the dog and gives further opportunity to assess the tissues while the dog is not bearing weight. Note the muscle tone and any swelling, masses, or temperature differences between body areas ([Figure 27-28](#)).

Effleurage

Effleurage is often performed to begin a massage, and is useful for relaxation, and to decrease swelling ([Box 27-4](#)). Beginning at a distal area, such as the paws, the hands move proximally, using medium pressure. This helps to move any toxins in the body toward the lymph nodes and aids drainage ([Figure 27-29](#)).

Petrissage

Petrissage is a type of massage with applied pressures which are deep and compress the underlying tissues. Kneading, wringing, skin rolling and pick-up-and-squeeze are petrissage movements.

Compressions

When performing effleurage, an increase in muscle tone in certain muscles may be identified. The triceps and deltoid muscles are commonly affected. Compressions to the affected muscles may be easily administered and are often